MOHIT SINGH RAJPUT

ML & Generative AI Engineer

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TECHNICAL SKILLS

Programming Languages: Python, C, C++, Java, JavaScript

ML / AI: Machine Learning, Deep Learning, Computer Vision, NLP, Generative AI, LLMs, RAG, Prompt Engineering

Frameworks, Libraries & Tools: TensorFlow, Keras, Scikit-learn, OpenCV, LangChain, LangGraph, NumPy, Pandas, Matplotlib, Seaborn, FastAPI, HTML, CSS, SQL

SUMMARY

ML & Generative AI Engineer with end-to-end experience across NLP, computer vision, and applied deep learning, seeking to apply expertise in LLM applications, RAG pipelines, and multi-agent systems to build tested, documented, and production-ready AI solutions.

EXPERIENCE

AI/ML Intern — Labmentix *June – September 2026*

Multi-Agent LLM Systems & Computer Vision | RAG, FastAPI, TensorFlow, TFLite

- Completed a 3-month internship (Jun 1 – Sep 1, 2026) focused on applied ML and deep learning projects.
- Built and evaluated two deep learning architectures (a from-scratch CNN and a fine-tuned MobileNetV2) for facial emotion recognition, covering data preprocessing, augmentation, training, and TFLite quantization for on-device deployment.
- Designed a multi-agent LLM customer support system with intent-based routing, RAG retrieval, and a FastAPI backend, deployed live across Render and Vercel.
- Applied consistent evaluation discipline across both projects, including held-out test sets, quantitative benchmarking of every model variant, and automated tests.

Machine Learning Intern — CodeAlpha *June 2026*

Project-Based ML Internship | Python, Scikit-learn, Pandas

- Completed a project-based machine learning internship covering end-to-end data preprocessing, feature engineering, and model evaluation.
- Applied supervised and unsupervised techniques (classification, clustering) to real-world datasets in Python with scikit-learn and pandas.

PROJECTS

Multi-Agent LLM Customer Support Platform | *Python, FastAPI, Next.js, FAISS, Sentence-Transformers, Groq/OpenAI, SQLAlchemy, JWT, Twilio, SendGrid* Live App

- Architected a 5-agent customer support system (billing, technical, product, complaints, FAQ) behind a two-stage intent router, reaching 90% intent-classification and agent-routing accuracy on a multilingual evaluation set.
- Built a RAG pipeline (FAISS, Sentence-Transformers) over an 8-document knowledge base, and a FastAPI backend with REST endpoints for auth, chat, analytics, and admin.
- Integrated Twilio (WhatsApp) and SendGrid (email) for automated ticket escalation, and deployed the full stack to production on Render and Vercel with retry logic for LLM call resilience.

Loan-Eligibility-and-EMI-Prediction-AI | *Python, Scikit-learn, XGBoost, MLflow, Streamlit, Pandas, Joblib* Live App

- Built a dual-target ML system on 404,800 loan-applicant records, classifying EMI eligibility (Eligible/High_Risk/Not_Eligible) and regressing the maximum safe monthly EMI from 42 engineered financial-ratio features.
- Benchmarked four classifiers and four regressors via GridSearchCV/RandomizedSearchCV with MLflow experiment tracking; selected XGBoost for both tasks, reaching 97.0% classification accuracy and 0.99 R2 on EMI regression.
- Shipped a 4-page Streamlit application for eligibility prediction, EMI calculation, model performance, and data insights, with models serialized via Joblib for deployment.

DeepFER — Facial Emotion Recognition | *Python, TensorFlow/Keras, OpenCV, TFLite, Flask, pytest* Live App

- Trained and benchmarked two CNN architectures for 7-class facial emotion recognition on FER-2013 (35,887 images), reaching 52.27% and 59.46% accuracy on a held-out test set.
- Quantized both models to TFLite (float32, int8, full int8); full-integer quantization cut latency by up to 108x with no accuracy loss.
- Shipped a Flask web app for static-image and live browser-camera inference, backed by an automated pytest suite and reproducible evaluation artifacts.

Tourism Experience Analytics System | *Python, Scikit-learn, LightGBM, XGBoost, Pandas, Streamlit, Joblib, sqlite3* Live App

- Built an end-to-end supervised learning system on 50,000+ tourism transactions spanning 9 relational tables, 33,530 users, and 30 attractions across 5 continents, with SQL-based exploration via 6 production-style queries (CTEs, window functions, subqueries).
- Benchmarked 11 regression and 11 classification algorithms; selected LightGBM (RMSE 0.72, R^2 0.34) for rating prediction and Extra Trees (51.2% weighted F1, stable across 5-fold CV) for visit-mode classification.
- Designed a hybrid recommender (item-based collaborative filtering + TF-IDF content-based) outperforming the popularity baseline by 4.8× on Precision@5 and Recall@5, deployed as an interactive Streamlit application.

EDUCATION

Bachelor of Technology (B.Tech), Computer Science & Engineering *Aug 2023 – Mar 2027 (Expected)*

Shankara Institute of Technology — Jaipur, Rajasthan

CERTIFICATES / EXTRACURRICULAR

- Oracle Certified Foundations Associate — Agentic AI (Oracle University, Aug 2026)
- Deloitte Australia Data Analytics Job Simulation — Forage (Aug 8, 2026)
- Tata Group Data Analytics Job Simulation — Forage (Aug 18, 2026)
- Walmart USA Advanced Software Engineering Virtual Experience Program — Forage (Aug 9, 2026)
- AWS APAC Solutions Architecture Virtual Experience Program — Forage (Aug 20, 2026)
- Tata Data Visualisation: Empowering Business with Effective Insights — Forage (Aug 8, 2026)
- Datacom Partnering with AI in the Workplace Job Simulation — Forage (Aug 16, 2026)